

STOR 155.010 Lecture 05

Visualizing Data

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Overview

1 Numerical Data

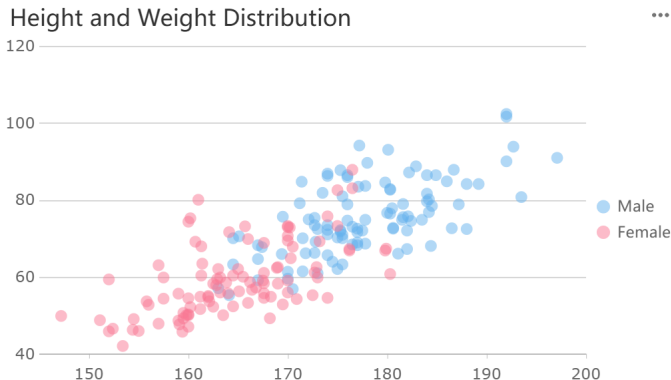
2 Categorical Data

Visualization of Numerical Data

- Scatterplot
- Dot plot
- Histogram
- Boxplot

Scatterplot

- Useful for visualizing the relationship between **two** numerical variables.
- They provide information on positive/negative association.



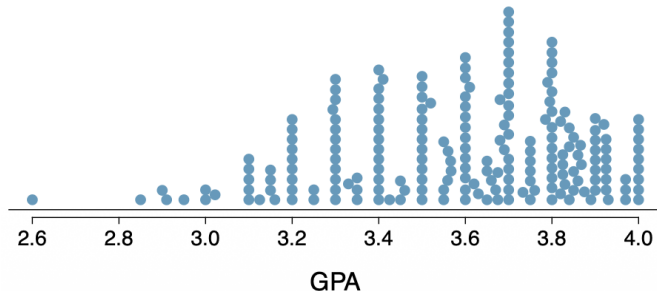
Dot plots

- Useful for visualizing **one** numerical variable.
- Darker colors represent areas where there are more observations.
- Indicate the mean of the data in the dot plot to give an idea about the center of the distribution.



Stacked Dot Plot

- Higher bars represent areas where there are more observations.
- Makes it a little easier to judge the center and the shape of the distribution.

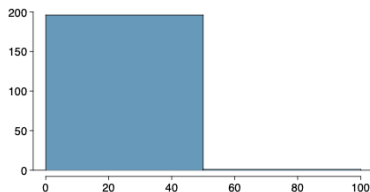


Histograms

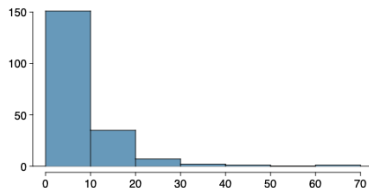
- Used to visualize **one** numerical variable.
- Rectangles whose **area** represents frequency; width represents a consecutive, non-overlapping interval.
- X-axis: possible values of the variable. Y-axis: frequency of each value.
- Horizontal range split into regular sub-ranges called **bins**.
- Height of each rectangle = number of data points in that bin.
- Great for spotting **skewness** and **kurtosis** in a large dataset.

Bin width

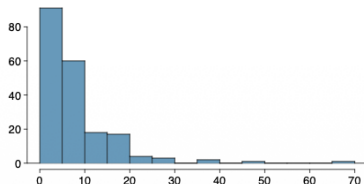
- The same dataset can lead to very different histograms based on the chosen bin width.
- Preferred: neither too many nor too few bins.



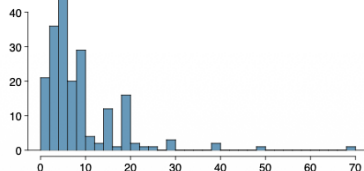
Hours / week spent on extracurricular activities



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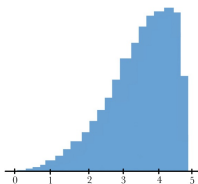


Hours / week spent on extracurricular activities

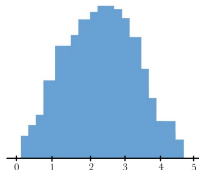


Hours / week spent on extracurricular activities

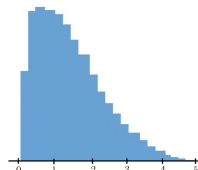
Skewness and Modality



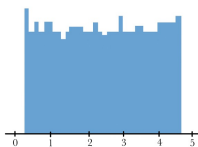
skew left



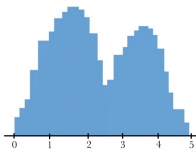
symmetric, unimodal



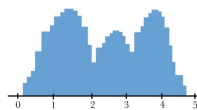
skew right



uniform

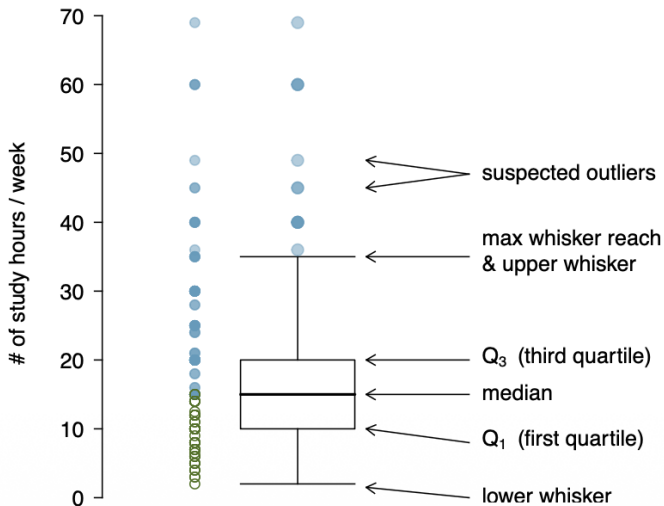


bimodal



multimodal

Box plot



Whiskers and outliers

- Define the **interquartile range** to be $IQR = Q_3 - Q_1$.
- **Whiskers** of a box plot can extend up to $1.5 \times IQR$ away from the quartiles.

$$\text{max upper whisker reach} = Q_3 + 1.5 \times IQR$$

$$\text{max lower whisker reach} = Q_1 - 1.5 \times IQR$$

$$IQR : 20 - 10 = 10$$

$$\text{max upper whisker reach} = 20 + 1.5 \times 10 = 35$$

$$\text{max lower whisker reach} = 10 - 1.5 \times 10 = -5$$

- A potential **outlier** is defined as an observation beyond the maximum reach of the whiskers. It is an observation that appears extreme relative to the rest of the data.

Median and IQR are more **robust** to skewness and outliers than mean and SD. Therefore,

- for skewed distributions it is often more helpful to use median and IQR to describe the center and spread
- for symmetric distributions it is often more helpful to use the mean and SD to describe the center and spread

Overview

1 Numerical Data

2 Categorical Data

Visualization of Categorical Data

- Contingency tables
- Bar plots
- Mosaic plots
- Pie charts
- Side-by-side box plots

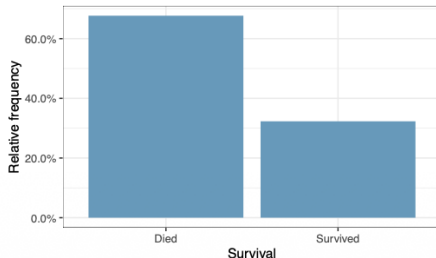
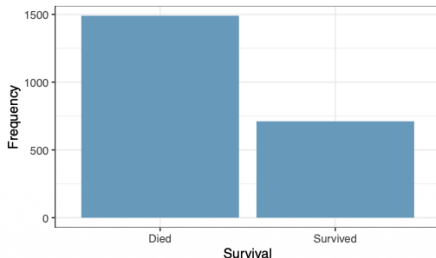
Contingency tables

- A table that summarizes data for two categorical variables is called a **contingency table**.
- The contingency table below shows the distribution of survival and ages of passengers on the Titanic.

		Survival		Total
		Died	Survived	
Age	Adult	1438	654	2092
	Child	52	57	109
	Total	1490	711	2201

Bar plots

- A **bar plot** is a common way to display a single categorical variable.
- A bar plot where proportions instead of frequencies are shown is called a **relative frequency bar plot**.



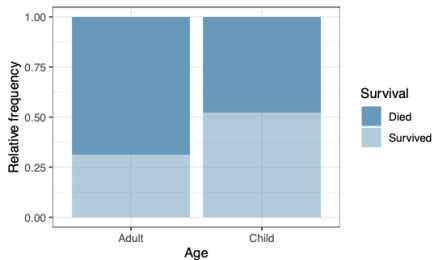
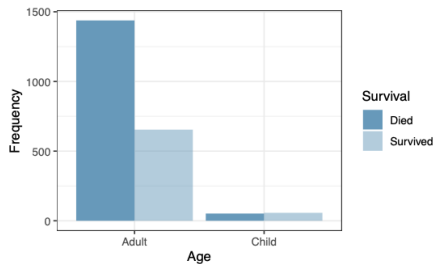
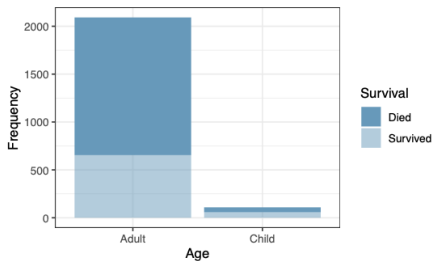
Bar plots vs. Histograms

- Bar plots are used for displaying distributions of categorical variables.
- Histograms are used for numerical variables.
- The x-axis in a histogram is a number line, hence the order of the bars cannot be changed.
- In a bar plot, the categories can be listed in any order (though some orderings make more sense than others, especially for ordinal variables.)

Bar plots with two variables

- **Stacked bar plot:** Graphical display of contingency table information, for counts.
- **Side-by-side bar plot:** Displays the same information by placing bars next to, instead of on top of, each other.
- **Standardized stacked bar plot:** Graphical display of contingency table information, for proportions.

Example



Mosaic Plots

- Graphical view of a **contingency table**.
- Area of each rectangle $\propto$ cell frequency.
- Width = one variable's proportions; height (within each column) = the other variable's proportions.
- Great for spotting **association** between two categorical variables — uneven splits across columns signal dependence.
- Color/shading can highlight a third variable, a test result, or a specific cell of interest.

Example

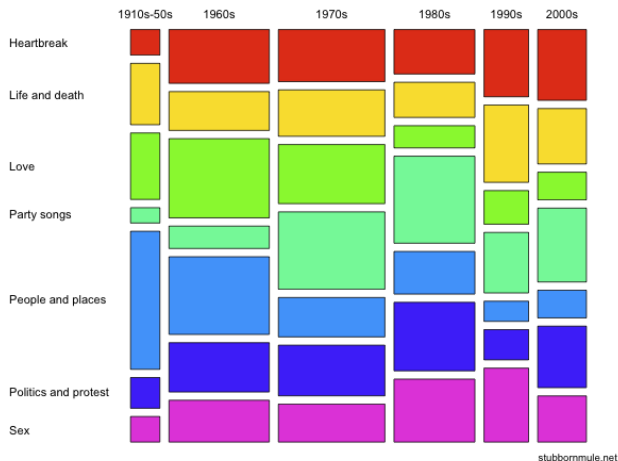
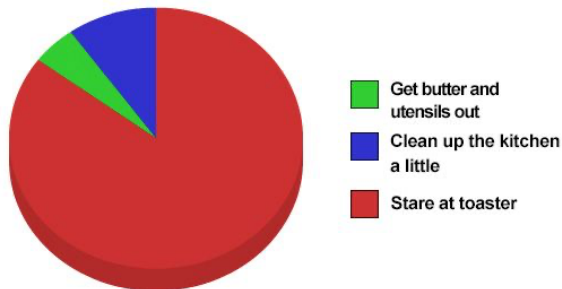


FIGURE: Cross-sectional distribution through time of different musical themes in the Guardian's list of "1000 songs to hear before you die".

Pie Charts

- Circular chart divided into slices.
- Slice angle (and area) $\propto$ category's share of the whole.
- Best for **one** categorical variable with few levels.
- Good for showing parts of a **whole** (percentages summing to 100%).
- Limitation: hard to compare slice sizes precisely — bar charts are often clearer for more than a handful of categories.

WHAT I DO WHILE WAITING FOR MY TOAST TO POP UP



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